

Saif Khan

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TECHNICAL SKILLS

Languages & Backend: Python, Flask, FastAPI, Pydantic, REST APIs, HTML, CSS

Data Science: NumPy, Pandas, Matplotlib, Seaborn, EDA, Data Preprocessing, Feature Engineering, Time Series

Machine Learning: Regression, Classification, Clustering, PCA, Model Evaluation, Cross-Validation, Hyperparameter Tuning, Ensemble Learning, Scikit-learn, XGBoost

Deep Learning: ANN, CNN, RNN, LSTM, GRU, Transformers

Generative AI & Agents: LLMs, Prompt Engineering, LangChain, RAG, AI Agents, LangGraph, Tool Calling, Tavily API, Human-in-the-Loop

NLP: NLTK, TF-IDF, Bag-of-Words

Databases & Development Tools: MySQL, Git, GitHub, Streamlit

EXPERIENCE

Data Science Intern · Suprmentr Technologies Pvt.Ltd.

Feb-May 2026

Engineering Employability Analytics: What Drives Graduate Success in India?  | Python · Pandas · XGBoost · SHAP · Streamlit

- Designed and developed an end-to-end employability analytics platform on **56,000+** engineering student records to investigate key drivers of graduate placement and salary outcomes.
- Engineered and evaluated an **XGBoost** prediction model (ROC-AUC **0.82**, 5-fold CV 0.81 ± 0.02), using **SHAP** explainability to reveal that internships and college tier influence employability more strongly than CGPA.
- Delivered actionable insights including a **42%** salary premium for Computer Science graduates and deployed a public-facing Streamlit dashboard for interactive career trend exploration.

PROJECTS

Where Does India's ₹33 Lakh Crore Tax Revenue Go?  | Python · Scikit-learn · Streamlit · NLP

- Led the development of a public-finance analytics platform by consolidating **18 years** of government taxation, expenditure, and state finance data into a unified analytical framework.
- Engineered fiscal performance indicators and applied K-Means clustering, Holt-Winters forecasting, and statistical analysis to identify revenue concentration, taxpayer participation trends, and state-level development patterns.
- Uncovered key insights including **60%** GST concentration across **six states** and taxpayer participation below **2%** of India's population, highlighting structural imbalances within the tax ecosystem.
- Designed and deployed an interactive Streamlit dashboard enabling exploration of **India's ₹33 lakh crore tax flow**, ministry spending efficiency, and long-term fiscal trends through data-driven visualizations.

How Global Oil Crises Impact India's Economy  | Python · Pandas · Time-series · nltk · arima · Streamlit

- Built an end-to-end economic intelligence platform using **six years** of geopolitical, crude oil, exchange rate, and trade datasets to analyze how global conflicts influence India's economic performance.
- Applied time-series forecasting, statistical modeling, and correlation analysis to quantify economic risks, finding that every **\$10** increase in crude oil prices **weakens** the **Indian Rupee** and increases annual import costs by approximately **\$15 billion**.
- Developed an ARIMA forecasting model (MAE: \$0.25/barrel) and deployed an interactive Streamlit application enabling exploration of oil-price scenarios, inflation risks, and trade-cost projections.

Crowd Safety & Pilgrim Forecasting for Hajj Operations  | Python · Pandas · Scikit-learn · Streamlit

- Designed and developed a predictive analytics platform leveraging **26 years** of Hajj and Umrah data to evaluate crowd safety risks and forecast future pilgrimage demand.
- Engineered features from pilgrim volume, incident, and environmental datasets, applying statistical analysis and machine learning techniques to identify crowd density and extreme heat as primary contributors to **safety-related events**.
- Built and compared Machine Learning and forecasting models including Random Forest, Gradient Boosting, and ARIMA; achieved best performance with Random Forest ($R^2 = 0.75$) and forecasted **~1.9 Million pilgrims** for 2026.
- Delivered insights through a deployed Streamlit dashboard that allows users to explore historical patterns, risk indicators, and demand forecasts for **large-scale pilgrimage events**.

EDUCATION

B.E. — Computer Science & Engineering, Navodaya Institute of Technology

May 2026